

FINISHED INTELLIGENCE REPORT

Brute Force RDP Campaign — Azure Virtual Machine Environment

Field	Value
Classification	TLP:GREEN — For Security Operations Use
Date	June 25, 2026
Report ID	IR-2026-001
Author	Roshanak Khodaparast, Security Engineer
Organization	Log(N) Pacific
Incident Type	Brute Force Credential Attack — RDP
Disposition	True Positive — Contained, No Compromise

SECTION 1 — KEY JUDGMENTS

We assess with HIGH CONFIDENCE that external threat actors conducted a sustained brute force campaign targeting internet-exposed RDP endpoints in the Log(N) Pacific environment. The campaign originated from 6 distinct IP addresses targeting 2 production virtual machines over a 5-hour window. No successful authentication was observed. Immediate containment via NSG hardening reduced RDP brute force incidents by 100% following implementation.

Recommended action for leadership: Enforce organizational policy requiring all Azure virtual machines to restrict RDP access to authorized IP ranges only — eliminating public internet exposure of remote management services.

SECTION 2 — THREAT OVERVIEW

Field	Value
Incident Type	Brute Force Credential Attack — RDP
Target	Azure VMs onboarded to Microsoft Defender for Endpoint

Timeframe	5-hour active campaign window
Detection	Microsoft Sentinel scheduled analytics rule — DeviceLogonEvents
ATT&CK Tactic	Credential Access (TA0006)
ATT&CK Technique	T1110.003 — Password Spraying
Kill Chain Phase	Exploitation (Stage 4)

Internet-exposed RDP (port 3389) is one of the most actively targeted attack surfaces in cloud environments. Automated scanning tools continuously probe Azure IP ranges for open RDP endpoints. A single successful authentication grants full remote desktop access — enabling lateral movement, data exfiltration, or ransomware deployment.

SECTION 3 — TECHNICAL ANALYSIS

3.1 Detection

A scheduled analytics rule was deployed in Microsoft Sentinel querying the DeviceLogonEvents table in Microsoft Defender for Endpoint:

```
DeviceLogonEvents
| where ActionType == "LogonFailed"
| where TimeGenerated > ago(5h)
| summarize EventCount = count()
  by RemoteIP, DeviceName
| where EventCount >= 10 | order by EventCount
```

Rule Setting	Value	Rationale
Run frequency	Every 4 hours	Timely detection
Lookback window	5 hours	Catches sustained campaigns
Alert grouping	Single incident / 24h	Reduces alert fatigue
Entity mappings	RemoteIP + DeviceName	Enables pivot investigation
Incident auto-create	Enabled	Immediate SOC notification

3.2 Attack Findings

Finding	Detail	Significance
Attacking IPs	6 distinct source IPs	Multiple origin — automated tool
Targeted hosts	2 Azure virtual machines	Opportunistic — not targeted

Failed logons	10+ per IP per 5h window	Threshold met — rule triggered
Successful logons	ZERO — confirmed	No compromise occurred
Protocol targeted	RDP — port 3389	Common attack surface

Successful logon verification query used during investigation:

```
let TargetDevice = "windows-target-1";
let SuspectIP = "89.116.158.44";
DeviceLogonEvents
| where ActionType == "LogonSuccess"
| where DeviceName == TargetDevice
| where RemoteIP == SuspectIP
| order by TimeGenerated desc // Result: ZERO rows — no successful authentication confirmed
```

3.3 ATT&CK Mapping

ATT&CK Element	ID	Detail
Tactic	TA0006	Credential Access
Technique	T1110	Brute Force
Sub-technique	T1110.003	Password Spraying
Persistence (if successful)	T1053.005	Scheduled Task
Lateral Movement (if successful)	T1021.001	Remote Desktop Protocol
Impact (if successful)	T1486	Data Encrypted for Impact

3.4 Threat Actor Assessment

Assessment	Confidence	Reasoning
Financially motivated	MEDIUM	RDP brute force = primary ransomware initial access
Automated scanning tool	HIGH	Multiple IPs, high volume, consistent pattern
Opportunistic not targeted	HIGH	Generic credential stuffing — no specific targeting
Nation-state unlikely	HIGH	TTPs inconsistent with APT tradecraft — too noisy

Attribution: Unattributed opportunistic threat actor using commodity RDP brute force tooling. Consistent with ransomware affiliate initial access operations or access broker activity.

SECTION 4 — RECOMMENDED ACTIONS

Priority	Action	Owner
P1 — COMPLETED	Lock down NSG to authorized IPs only	Security Engineering
P1 — COMPLETED	Verify zero successful authentications	SOC Analyst
P2 — SHORT TERM	Enforce MFA on all RDP-accessible accounts	Identity Team
P2 — SHORT TERM	Azure Policy — restrict RDP org-wide	Cloud Security
P2 — SHORT TERM	Enable JIT VM access in Defender for Cloud	Cloud Security
P3 — STRATEGIC	Replace RDP with Azure Bastion or VPN	Architecture
P3 — STRATEGIC	Account lockout after 5 failed attempts	Identity Team

Enhanced detection query — catches successful brute force (active compromise indicator):

```
let BruteForceIPs = DeviceLogonEvents
| where ActionType == "LogonFailed"
| where TimeGenerated > ago(5h)
| summarize FailCount = count()
  by RemoteIP, DeviceName
| where FailCount >= 10;
DeviceLogonEvents
| where ActionType == "LogonSuccess"
| join kind=inner BruteForceIPs on RemoteIP, DeviceName
| project TimeGenerated, DeviceName, RemoteIP, AccountName, FailCount // This catches
worst case – successful login after brute force = active compromise
```

SECTION 5 — INDICATORS OF COMPROMISE

IOC Type	Value	Context
Remote IP	89.116.158.44	Observed brute force source
Log source	DeviceLogonEvents	Detection table in Defender/Sentinel
Malicious ActionType	LogonFailed (10+ per 5h)	Brute force indicator
Compromise indicator	LogonSuccess after	Active compromise if observed

	LogonFailed	
Target protocol	RDP — port 3389	Attack surface exploited
Impacted assets	2 Azure VMs	Targeted endpoints

SECTION 6 — INCIDENT TIMELINE

Field	Value
T+00:00	First failed RDP logon attempts detected — automated scanning begins from 6 IPs against 2 target VMs
T+05:00	Sentinel analytics rule triggers — 10+ failures per IP per 5-hour window met — incident auto-created
T+05:15	Incident assigned to analyst — status set to Active — investigation initiated per NIST 800-61
T+05:30	Entity mapping reviewed — 6 source IPs, 2 target hosts identified and documented
T+05:45	Successful logon verification completed — ZERO successful authentications confirmed
T+06:00	Containment applied — NSG locked down to authorized IPs only — public RDP exposure eliminated
T+06:30	Incident closed as True Positive — no compromise confirmed — policy recommendation documented

SECTION 7 — LESSONS LEARNED

Lesson	Finding	Action Taken
Public RDP exposure	High-risk attack surface — exploited immediately	NSG hardening now standard
Detection efficacy	Sentinel rule fired within 5-hour window	Rule validated — no changes needed
Verification discipline	Must confirm no success before closing	Verification query added to playbook
Policy gap	No existing corporate NSG policy	Azure Policy enforcement proposed
Detection gap	Rule detects attempts, not success	Enhanced query built and documented

Incident Disposition: TRUE POSITIVE — CONTAINED

Analyst: Roshanak Khodaparast | Status: CLOSED | No Compromise Confirmed